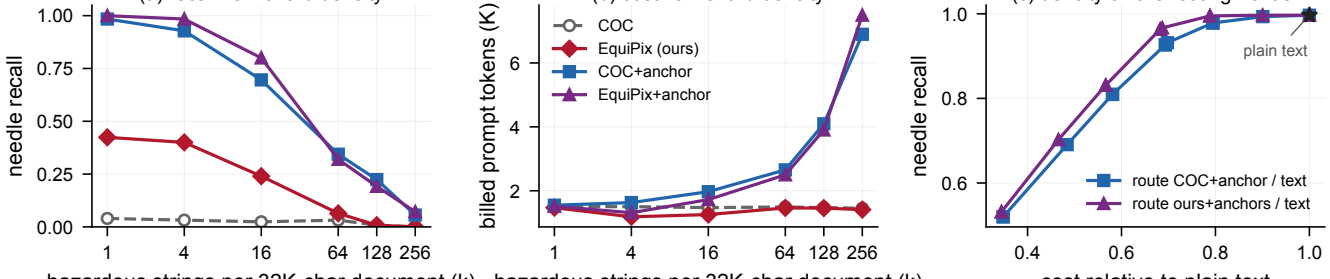
(a) recall vs. hazard density
needle recall
1.00
0.75
0.50
0.25
0.00
1
4
16
64
128
256
hazardous strings per 32K-char document (k)
(b) cost vs. hazard density
billed prompt tokens (K)
6
4
2
COC
EquiPix (ours)
COC+anchor
EquiPix+anchor
1
4
16
64
128
256
hazardous strings per 32K-char document (k)
(c) density-aware routing frontier
needle recall
1.0
0.8
0.6
plain text
route COC+anchor / text
route ours+anchors / text
0.4
0.6
0.8
1.0
cost relative to plain text